

# BHMS

## Boutique Hotel Management System

### Phase 2 — Complete Documentation

#### FINAL VERSION

CS251 Software Engineering 1 | Capital University | Spring 2025-2026

## PART 1 — Updated Software Requirements Specification

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□ All 6 Phase 1 deliverables have been updated to their final versions per Phase 2 requirements.

### 1.1 Purpose

This SRS defines the complete software requirements for the BHMS — a web-based Property Management System for boutique hotels, built using PHP (OOP), MySQL, MVC pattern, and Bootstrap. It follows IEEE 830 and the CS251 project template requirements.

□ *Rationale: A formal SRS prevents ambiguity, provides a shared baseline for all 7 team members, and enables systematic validation and testing.*

### 1.2 System Scope

The BHMS integrates four core modules:

- Front-Desk & Reservation: availability calendar, check-in/out state machine, dynamic room allocation, walk-in pricing
- Guest Profile & CRM: preferences, loyalty tiers, VIP flagging, GDPR anonymization, feedback loop
- Housekeeping & Maintenance: real-time room status machine, task assignment, inspection scoring, work-order escalation
- Billing & Ancillary: folio consolidation, POS bridge, multi-currency settlement, night audit simulator, RBAC

The system will NOT:

- Process real-time online payments (simulation only)
- Integrate with physical hotel hardware
- Provide a native mobile application (web-based only)

### 1.3 Glossary and Abbreviations

| Term         | Definition                                                                    |
|--------------|-------------------------------------------------------------------------------|
| Reservation  | Booking a room for a specific future period (Inquiry → FolioClosed)           |
| Check-in     | Guest arrival: identity verification, room assignment, digital key generation |
| Check-out    | Guest departure: billing settlement, room release, HK task creation           |
| Housekeeping | Department managing cleaning, inspection, and room status transitions         |
| VIP Guest    | Guest with loyaltyPoints ≥ 500 OR totalSpend ≥ EGP 10,000                     |
| Folio        | Master record of all financial transactions for one guest stay                |
| Rack Rate    | Walk-in price = basePrice × occupancyFactor (no advance booking discount)     |
| PMS          | Property Management System — the complete hotel platform                      |
| RBAC         | Role-Based Access Control — UI and data filtered by staff role                |
| GDPR         | Right to be Forgotten: PII anonymized, financial records preserved            |
| OOP          | Object-Oriented Programming — all PHP classes follow OOP principles           |
| MVC          | Model-View-Controller — mandatory architectural pattern                       |
| CCM          | Cyclomatic Complexity Metric = decisions + 1                                  |
| SSD          | System Sequence Diagram — actor ↔ system black-box interaction                |

### 1.4 System Stakeholders

| Stakeholder   | Role                                         | Primary Concern            | Key UCs                    |
|---------------|----------------------------------------------|----------------------------|----------------------------|
| Guest         | Books rooms, updates profile, gives feedback | Convenience, privacy       | UC-02, UC-12, UC-21        |
| Receptionist  | Check-in/out, billing, room allocation       | Speed, accuracy            | UC-01, UC-02, UC-33, UC-36 |
| Housekeeper   | Cleans rooms, updates status, logs minibar   | Clear tasks, simple UI     | UC-23, UC-24, UC-30, UC-32 |
| Supervisor    | Inspects rooms, manages HK staff             | Quality scoring, oversight | UC-24, UC-29               |
| Manager       | Reports, pricing, staff, overbooking control | Revenue, performance       | UC-03, UC-05, UC-38        |
| Accountant    | Folios, payments, financial reports          | Accuracy, audit trail      | UC-33, UC-36, UC-41        |
| Night Auditor | Runs end-of-day audit simulation             | Safe simulation, accuracy  | UC-42                      |
| Admin         | Users, roles, GDPR, system config            | Security, RBAC, compliance | UC-40, UC-19, UC-41        |

## 2. Functional Requirements

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□ Updated in this final version: UC numbering unified across all documents. Mathematical specifications added. Rationale provided for every requirement.

## 2.1 Requirements Notation (4 Types — as required by template)

All requirements are expressed using all four required notations:

- Natural Language: prose descriptions below
- Structured Natural Language: use-case templates (UC-ID, Goal, Actor, Pre/Post, Scenarios)
- Graphical Notations: Use-Case Diagrams, Activity Diagrams (see attached)
- Mathematical Specification: formal formulas in Section 2.3

## 2.2 Core CRUD Operations

The system shall support full CRUD for:

| Entity         | Create                | Read                   | Update                      | Delete / Anonymize                |
|----------------|-----------------------|------------------------|-----------------------------|-----------------------------------|
| Guest Profile  | Add new guest         | Search, view history   | Update PII, VIP status      | GDPR anonymization (soft delete)  |
| Room Inventory | Add room              | View floor map, status | Update status, key          | Mark OutOfOrder                   |
| Staff Accounts | Create staff          | List staff, view roles | Change role, reset password | Disable account (soft)            |
| Reservations   | Create booking        | Search, view lifecycle | State transition, reassign  | Cancel (state change, not delete) |
| HK Tasks       | Auto or manual create | View task list         | Assign, status update       | Mark Skipped                      |
| Services       | Add service           | View catalog           | Update price, active flag   | Deactivate                        |
| Folio Charges  | Post charge           | View breakdown         | Manual override (manager)   | Credit/refund only                |

## 2.3 Mathematical Specification

The following formulas define core algorithmic behaviors:

### Room Allocation Score (UC-01)

```
Score(room, guest) =  $\sum w_i \times match_i(\text{room.attribute}_i, \text{guest.preference}_i)$ 
where:  $w_i \in \{0.1, 0.2, 0.3\}$  = preference weight
       $match_i \in \{0, 1\}$  = exact match indicator
BestRoom = argmax { Score(roomk, guest) } for all k where roomk.status = 'Ready'
```

□ Rationale: A scoring algorithm prevents biased manual assignment and maximizes guest satisfaction measurably.

### Walk-In Pricing / Rack Rate (UC-05)

```
RackRate = BasePrice × OccupancyFactor × SeasonFactor
```

```

OccupancyFactor = 1 + (currentOccupancy% / 100) × 0.20    [if occupancy > 80%]
OccupancyFactor = 1.00                                   [if occupancy ≤ 80%]
SeasonFactor ∈ {0.90, 1.00, 1.25}    [Low, Standard, Peak]

```

□ *Rationale: Dynamic pricing maximizes revenue during high-demand periods and improves occupancy during low-demand periods.*

### VIP Threshold (UC-14)

```

VIPStatus = true  iff  LoyaltyPoints ≥ 500  OR  TotalLifetimeSpend ≥ EGP 10,000
LoyaltyTier = { Bronze   : 0 ≤ TotalNights < 5
                { Silver  : 5 ≤ TotalNights < 15
                { Gold    : 15 ≤ TotalNights < 30
                { Platinum: TotalNights ≥ 30

```

□ *Rationale: Objective, formula-based VIP criteria eliminate subjectivity and ensure consistent treatment.*

### Folio Total (UC-33)

```

Subtotal = Σ chargei  for all i in folio_charges WHERE folio_id = F
VAT = Subtotal × 0.14      (Egyptian VAT = 14%)
FolioTotal = Subtotal + VAT - DepositPaid
BalanceDue = max(0, FolioTotal - Σ payments_completed)

```

□ *Rationale: Automated tax calculation prevents manual errors and ensures legal compliance with Egyptian tax law.*

### Multi-Currency Settlement (UC-36)

```

ForeignAmount = TotalEGP / ExchangeRate(currency, timestamp)
Rate valid if: (NOW() - rate.fetchedAt) < 300 seconds (5 min window)
If rate expired: useCachedRate() AND notifyManager()

```

□ *Rationale: Boutique hotels in Egypt serve international guests; automated conversion prevents calculation errors at checkout.*

### Cyclomatic Complexity Threshold

```

M = E - N + 2P    (E=edges, N=nodes, P=connected components)
Equivalently: M = number_of_decisions + 1
Acceptable range: M ≤ 10. Refactoring required if M > 10.

```

## 2.4 Complete Use-Case Master Table (Corrected & Final)

□ *Numbering unified with Activity Diagrams, Sequence Diagrams, Collaboration Diagrams, and all Phase 2 deliverables.*

| UC-ID | Name                             | Module     | Actor        | Priority  | Rationale (Why Required)                                                     |
|-------|----------------------------------|------------|--------------|-----------|------------------------------------------------------------------------------|
| UC-01 | Dynamic Room Allocation Engine   | Front-Desk | Receptionist | Must Have | Prevents biased manual assignment; maximizes guest satisfaction              |
| UC-02 | Multi-State Reservation Workflow | Front-Desk | Receptionist | Must Have | State machine prevents invalid transitions; audit trail for legal compliance |

| UC-ID | Name                               | Module     | Actor        | Priority    | Rationale (Why Required)                                                  |
|-------|------------------------------------|------------|--------------|-------------|---------------------------------------------------------------------------|
| UC-03 | Overbooking Risk Controller        | Front-Desk | Manager      | Should Have | Reduces revenue loss from empty rooms using historical cancellation rates |
| UC-04 | Early Check-In Eligibility Checker | Front-Desk | Receptionist | Should Have | Cross-references HK status; prevents assigning unready rooms              |
| UC-05 | Walk-In Pricing Engine             | Front-Desk | Receptionist | Must Have   | Dynamic rack rates maximize revenue without prior reservation             |
| UC-06 | No-Show Penalty Trigger            | Front-Desk | System       | Must Have   | Automated penalty enforces booking commitments; releases room             |
| UC-07 | Digital Key Generator              | Front-Desk | System       | Could Have  | Time-limited encrypted key improves guest experience                      |
| UC-08 | Group Booking Coordinator          | Front-Desk | Receptionist | Could Have  | Single master account simplifies multi-room corporate bookings            |
| UC-09 | Deposit & Pre-Authorization        | Front-Desk | Receptionist | Should Have | Holds incidental amount; reduces financial risk for hotel                 |
| UC-10 | Checkout Queue Optimizer           | Front-Desk | System       | Could Have  | Staggered checkouts reduce lobby congestion                               |
| UC-11 | Automatic Room Upgrade Logic       | Front-Desk | System       | Should Have | Improves VIP satisfaction when booked category is unavailable             |
| UC-12 | Guest Preference Logger            | Guest CRM  | Staff        | Must Have   | Personalized service is core differentiator for boutique hotels           |
| UC-13 | Stay-History Aggregator            | Guest CRM  | Front Desk   | Must Have   | Lifetime value calculation drives loyalty tier decisions                  |
| UC-14 | Automated VIP Flagging             | Guest CRM  | System       | Must Have   | Objective criteria ensure consistent VIP treatment                        |
| UC-15 | Preference-to-HK Bridge            | Guest CRM  | System       | Should Have | Converts guest profile into actionable HK setup tasks                     |
| UC-16 | Pre-Arrival Concierge Wizard       | Guest CRM  | Guest        | Should Have | 48h pre-arrival form improves personalization at check-in                 |
| UC-17 | Blacklist & Security Flagging      | Guest CRM  | Manager      | Must Have   | Protects staff and property from known problematic guests                 |
| UC-18 | Anniversary/Birthday Logic         | Guest CRM  | System       | Could Have  | Surprise perks increase guest loyalty and positive reviews                |
| UC-19 | Privacy & Right to be Forgotten    | Guest CRM  | Admin        | Must Have   | GDPR compliance is legally mandatory;                                     |

| UC-ID | Name                               | Module       | Actor         | Priority    | Rationale (Why Required)                                            |
|-------|------------------------------------|--------------|---------------|-------------|---------------------------------------------------------------------|
|       |                                    |              |               |             | financial records preserved                                         |
| UC-20 | Referral Reward Tracking           | Guest CRM    | Guest         | Could Have  | Referral credits incentivize organic marketing                      |
| UC-21 | Post-Stay Feedback Loop            | Guest CRM    | Guest         | Should Have | Structured feedback linked to on-duty staff drives improvement      |
| UC-22 | Corporate Account Manager          | Guest CRM    | Sales Manager | Could Have  | Central billing simplifies corporate client relationship management |
| UC-23 | Real-Time Room Status Machine      | Housekeeping | System        | Must Have   | State machine prevents booking an unclean room                      |
| UC-24 | Housekeeper Assignment Logic       | Housekeeping | Supervisor    | Must Have   | Dynamic workload balancing improves cleaning efficiency             |
| UC-25 | Maintenance Work-Order Escalation  | Housekeeping | Staff         | Must Have   | Critical issues automatically trigger OOO status to protect guests  |
| UC-26 | Linen & Consumable Inventory Sync  | Housekeeping | Housekeeper   | Should Have | Real-time deduction prevents stock-outs mid-shift                   |
| UC-27 | Lost and Found Management          | Housekeeping | Housekeeper   | Should Have | Secure log linked to guest profile enables fast reunification       |
| UC-28 | Preventative Maintenance Scheduler | Housekeeping | Manager       | Should Have | Scheduled deep-cleans extend asset life and prevent breakdowns      |
| UC-29 | Inspection Quality Scoring         | Housekeeping | Supervisor    | Should Have | Objective scoring drives housekeeper performance improvement        |
| UC-30 | Minibar Consumption Logger         | Housekeeping | Housekeeper   | Must Have   | Instant folio posting prevents revenue leakage at checkout          |
| UC-31 | Turn-Down Service Coordinator      | Housekeeping | Supervisor    | Could Have  | Evening room prep is a key boutique hotel differentiator            |
| UC-32 | HK-to-Front-Desk Instant Alert     | Housekeeping | Housekeeper   | Must Have   | Real-time alerts enable immediate response to priority arrivals     |
| UC-33 | Folio Consolidation Engine         | Billing      | Receptionist  | Must Have   | Single unified invoice eliminates manual aggregation errors         |
| UC-34 | External Service POS Bridge        | Billing      | POS Operator  | Should Have | Direct folio posting prevents lost charges from spa/café            |
| UC-35 | Service Cancellation Fee Logic     | Billing      | Receptionist  | Must Have   | Time-based penalties enforce cancellation policy compliance         |

| UC-ID | Name                         | Module  | Actor         | Priority    | Rationale (Why Required)                                           |
|-------|------------------------------|---------|---------------|-------------|--------------------------------------------------------------------|
| UC-36 | Multi-Currency Settlement    | Billing | Receptionist  | Must Have   | International guests require foreign currency settlement           |
| UC-37 | Shared Expense Split-Bill    | Billing | Receptionist  | Should Have | Multiple guests sharing a room need separate invoices              |
| UC-38 | Automatic Tax & Tourism Levy | Billing | System        | Must Have   | Egyptian VAT (14%) must be calculated and displayed on every folio |
| UC-39 | Pro-Forma Invoice Generator  | Billing | Billing Clerk | Should Have | Preview bill reduces disputes at checkout                          |
| UC-40 | Role-Based Access Control    | System  | Admin         | Must Have   | Data security requires role-filtered access to all modules         |
| UC-41 | System Audit Trail           | System  | Auditor       | Must Have   | Immutable log is legally required for financial audit compliance   |
| UC-42 | Night Audit Simulator        | System  | Night Auditor | Must Have   | Safe simulation finds discrepancies before daily close             |

## 3. Non-Functional Requirements (Final Version)

### 3.1 NFR Categories Used

Categories: Performance, Security, Usability, Scalability, Reliability, Maintainability, Portability

### 3.2 NFR Specification with Fit Criteria

| ID     | Requirement       | Category    | Fit Criteria (Testable)                                | Arch. Impact                             | Rationale                                                          |
|--------|-------------------|-------------|--------------------------------------------------------|------------------------------------------|--------------------------------------------------------------------|
| NFR-01 | Fast Page Loading | Performance | All pages load $\leq 2$ sec under 100 concurrent users | DB indexing, query optimization          | Booking pages > 2s cause 40% abandonment rate (industry benchmark) |
| NFR-02 | Data Encryption   | Security    | All PII encrypted at rest (bcrypt); HTTPS enforced     | password_hash(PASSWORD_DEFAULT); SSL/TLS | Unencrypted guest PII violates Egyptian data protection law        |

| ID     | Requirement        | Category        | Fit Criteria (Testable)                                     | Arch. Impact                                   | Rationale                                                    |
|--------|--------------------|-----------------|-------------------------------------------------------------|------------------------------------------------|--------------------------------------------------------------|
| NFR-03 | Staff Usability    | Usability       | New staff completes core task within 15 min first login     | Role-filtered sidebar; consistent UI           | High staff turnover in hotels requires minimal training time |
| NFR-04 | Concurrent Users   | Scalability     | 1000 simultaneous sessions without degradation              | Singleton DB; stateless PHP sessions           | Peak season occupancy requires multiple simultaneous users   |
| NFR-05 | System Uptime      | Reliability     | 99.9% uptime during business hours (max 8.7h downtime/year) | DB backups every 6h; fault tolerance           | Hotels operate 24/7; downtime directly loses revenue         |
| NFR-06 | Code Quality       | Maintainability | PSR-12 PHP standards; max CCM ≤ 10 per function             | MVC; modular service classes; Strategy pattern | Maintainable code reduces future bug-fix cost by ~40%        |
| NFR-07 | UI Consistency     | Usability       | Unified design system across all 11 modules                 | Central CSS variables; Bootstrap grid          | Inconsistent UI increases cognitive load and training time   |
| NFR-08 | Audit Immutability | Security        | No log entry modifiable by any user role                    | audit_logs: no UPDATE/DELETE permissions       | Tamper-proof logs required for financial audit compliance    |
| NFR-09 | GDPR Compliance    | Security        | PII anonymized within 24h of verified request               | Anonymization replaces PII with ANON_timestamp | GDPR Article 17 right to erasure is legally mandatory        |
| NFR-10 | Portability        | Portability     | Runs on PHP 8.0+ / MySQL 5.7+ on Windows/Linux/macOS        | No OS-specific code; XAMPP compatible          | Development on Windows, deployment on Linux is common setup  |

### 3.3 NFR Impact on Architecture

- NFR-04 + NFR-01 → Singleton Pattern (DatabaseConnection) prevents connection pool exhaustion under high concurrent load



- NFR-02 + NFR-08 → RBAC enforced at Auth layer; AuditLogger::log() called on every state-changing operation
- NFR-06 → MVC separates concerns; Strategy Pattern allows pricing rule changes without touching PricingEngine core
- NFR-09 → anonymizePII() replaces {name, email, phone, dob, passport} with ANON values while preserving folios/payments tables

## 4. Design & Implementation Constraints

| Constraint      | Specification                                                          | Rationale                                                     |
|-----------------|------------------------------------------------------------------------|---------------------------------------------------------------|
| Language        | PHP 8.0+ (OOP — classes, interfaces, inheritance, polymorphism)        | Required by CS251 technical specification                     |
| Architecture    | MVC Pattern (Model-View-Controller) — mandatory                        | Enforces separation of concerns; testability                  |
| Database        | MySQL via phpMyAdmin; MySQLi or PDO; 19 tables + 3 views               | Standard for XAMPP; relational integrity via FK constraints   |
| Frontend        | Bootstrap 5 — no ready-made templates; custom CSS variables            | Custom design required; no plagiarism from templates          |
| Design Patterns | Singleton (DB), Observer (Room status), Strategy (Pricing) — mandatory | 3 patterns required; Singleton explicitly required by spec    |
| AJAX            | Minimum 1 AJAX call (ajax_room_status.php — real-time room update)     | Required by CS251 technical specification                     |
| Authentication  | PHP Sessions + password_hash(PASSWORD_DEFAULT) = bcrypt                | Industry-standard password security                           |
| File Upload     | PHP file upload for guest documents / report export                    | Required by CS251 technical specification                     |
| Framework       | PHP Native preferred; Laravel permitted but not mandatory              | PHP Native for academic evaluation; no hidden framework magic |

## 5. System Evolution & Anticipated Changes

| Anticipated Change                 | Probability | Impact on Design      | Design Strategy                                                            |
|------------------------------------|-------------|-----------------------|----------------------------------------------------------------------------|
| Mobile app for guest self check-in | High        | New client layer      | API already HTTP/JSON; add Mobile UI Controllers without touching services |
| AI chatbot for 24/7 concierge      | Medium      | New NLP service class | Strategy Pattern allows plugging AIConciergeStrategy alongside existing    |

| Anticipated Change                     | Probability | Impact on Design                   | Design Strategy                                                |
|----------------------------------------|-------------|------------------------------------|----------------------------------------------------------------|
| GDS integration (Booking.com, Expedia) | High        | External booking API               | Adapter class in Service Layer; domain model unchanged         |
| Contactless payment gateway            | High        | Payment class extended             | Observer on Payment notifies GatewayService; no core change    |
| Multi-property (chain) management      | Medium      | property_id FK added to all tables | RBAC updated with property scope; tenant isolation at DB level |
| Tax rate changes                       | High        | Tax calculation update             | pricing_rules table already designed; zero code change needed  |
| GDPR updates                           | Medium      | Anonymization scope changes        | anonymizePII() method isolated; easy to extend fields          |

## 6. Requirements Discovery Approaches

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### 6.1 Simulated Stakeholder Interviews (Detailed Examples)

#### Interview 1 — Receptionist (Sara, 3 years experience):

- Q: What slows you down at check-in? → A: Finding the right room manually based on guest notes takes 5-10 minutes
- Discovery: UC-01 (Dynamic Room Allocation) with scoring algorithm needed
- Q: How do you handle foreign guests paying in USD? → A: We calculate manually with a phone calculator
- Discovery: UC-36 (Multi-Currency Settlement) with live exchange rate API

#### Interview 2 — Night Auditor (Alex):

- Q: What's your biggest risk in the night audit? → A: If I make a mistake, it affects live data and I can't undo it
- Discovery: UC-42 Night Audit Simulator with isSimulation=true flag — NEVER touches live data

#### Interview 3 — Housekeeper (Omar):

- Q: How do you notify front desk a priority room is ready? → A: I walk down to reception or call — takes too long
- Discovery: UC-32 (HK-to-Front-Desk Instant Alert) with real-time push notification

### 6.2 Existing System Analysis

- Cloudbeds (<https://cloudbeds.com>): identified room status state machine pattern (6 states: Clean→Occupied→Dirty→InCleaning→Inspecting→Ready)
- Mews (<https://mews.com>): identified digital key simulation, pre-arrival concierge wizard concept, loyalty tier structure
- Opera PMS: identified folio consolidation pattern, multi-currency settlement workflow, night audit simulation concept

## 6.3 Scenario-Based Analysis (Activity Diagrams)

- 7 swimlane Activity Diagrams created — one per team member — covering critical business processes
- AD-01: Room Allocation → discovered need for VIP priority lane and upgrade fallback
- AD-02: Reservation Workflow → discovered 5 valid state transitions + 2 alternative paths (NoShow, Cancel)
- AD-03: Night Audit → discovered isSimulation safety flag requirement
- AD-04: Multi-Currency → discovered 5-minute rate expiry window requirement
- AD-05: Housekeeping → discovered auto-create HK task on checkout requirement
- AD-06: Walk-In Pricing → discovered occupancy surge pricing (+20% when >80%)
- AD-07: HK Alert → discovered priority categorization (High → Critical alert, Standard → notification)

## 7. Requirements Validation Techniques

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### 7.1 Prototyping (PHP Implementation)

A working 20-file PHP prototype was built covering all 42 UCs. Validation: traced each UC through the implementation and verified output matches SRS post-conditions.

- UC-01 validated: scoring algorithm assigns room with highest match score ✓
- UC-42 validated: isSimulation=true flag prevents any write to live tables ✓
- UC-19 validated: anonymizePII() sets name=ANON\_timestamp, email=del@del.local, financial records unchanged ✓

### 7.2 Reviews and Inspections

- Consistency check: UC numbering unified across SRS, Activity Diagrams (7), Sequence Diagrams (12), Collaboration Diagrams (12), Use-Case Diagram, Class Diagram v1/v2/v3, Object Diagrams (7)
- Completeness check: all 42 functions have User Requirements, System Requirements, MoSCoW priority, and Rationale
- Traceability: each UC maps to at least one Activity Diagram, one SD, one Collaboration Diagram, one DB table, and one PHP page

### 7.3 Mathematical Verification (Detailed Examples)

**Folio calculation test:**

- Input: charges = EGP 500 (room) + EGP 250 (spa) = EGP 750
- VAT =  $750 \times 0.14$  = EGP 105 → FolioTotal = EGP 855
- Deposit paid = EGP 200 → BalanceDue =  $855 - 200$  = EGP 655 ✓

**VIP test:**

- Guest A: loyaltyPoints = 620 → VIPStatus = true ( $\geq 500$ ) ✓
- Guest B: totalSpend = EGP 12,000 → VIPStatus = true ( $\geq 10,000$ ) ✓
- Guest C: nights = 18 → LoyaltyTier = Gold ( $15 \leq 18 < 30$ ) ✓

**Walk-in pricing test:**

- BasePrice = EGP 800, occupancy = 85% → OccupancyFactor =  $1 + 0.20$  = 1.20

- $\text{RackRate} = 800 \times 1.20 = \text{EGP } 960 \checkmark$

#### Currency test:

- $\text{TotalEGP} = 960, \text{ExchangeRate(USD)} = 48.50 \rightarrow \text{ForeignAmount} = 960 / 48.50 = \$19.79 \checkmark$

## 7.4 Black-Box Boundary Testing (Sample)

| Function           | Boundary Test             | Input                              | Expected Output                       |
|--------------------|---------------------------|------------------------------------|---------------------------------------|
| Guest Registration | Empty name                | name="", email='x@x.com'           | Validation error: name required       |
| Payment Processing | Zero amount               | amount=0.00                        | Error: amount must be > 0             |
| Payment Processing | Exact balance             | amount=655.00 (=BalanceDue)        | Payment accepted, folio Closed        |
| Night Audit        | Folio diff = 0.01         | diff=0.01                          | NOT flagged ( $\leq$ threshold)       |
| Night Audit        | Folio diff = 0.02         | diff=0.02                          | Flagged as discrepancy (> threshold)  |
| HK Task Priority   | priority=5                | priority=5 (max is 4)              | Validation error: max priority is 4   |
| Reservation Date   | Same-day checkout         | checkin=today, checkout=today      | Error: checkout must be after checkin |
| GDPR Anonymize     | Active reservation exists | guestID with CheckedIn reservation | Error: cannot anonymize active guest  |

## Appendix A — User Interfaces (from Activity Diagrams)

| Page            | File             | Primary Users            | Key Functions                                         |
|-----------------|------------------|--------------------------|-------------------------------------------------------|
| Login           | login.php        | All staff                | Authentication, session, RBAC init                    |
| Dashboard       | dashboard.php    | Manager, Receptionist    | Live KPIs, occupancy, alerts, departures              |
| Reservations    | reservations.php | Receptionist             | CRUD, state transitions, folio link                   |
| Check-In/Out    | checkin.php      | Receptionist             | Arrivals/departures today, digital key generation     |
| Guest Profiles  | guests.php       | Receptionist, Concierge  | Profile CRUD, preferences, blacklist, GDPR            |
| Preferences     | preferences.php  | Receptionist, Concierge  | Add/edit/delete preferences by type, grouped by guest |
| Room Status Map | rooms.php        | Receptionist, Supervisor | Floor map, AJAX status update, add room               |
| Housekeeping    | housekeeping.php | Housekeeper, Supervisor  | Task create/assign/complete, inspection score         |
| Maintenance     | maintenance.php  | Staff, Manager           | Work orders, assign, escalate, resolve                |

| Page            | File            | Primary Users              | Key Functions                                     |
|-----------------|-----------------|----------------------------|---------------------------------------------------|
| Services        | services.php    | Receptionist, POS Operator | Service catalog, POS bridge (post to room)        |
| Inventory       | inventory.php   | Housekeeper, Supervisor    | Stock levels, restock, deduct, low-stock alerts   |
| Folio & Billing | folios.php      | Receptionist, Accountant   | Charges, multi-currency payment, close folio      |
| Feedback        | feedback.php    | Receptionist, Concierge    | Star rating, distribution chart, negative alerts  |
| Staff & RBAC    | staff.php       | Admin, Manager             | Add/disable staff, change roles, reset password   |
| Night Audit     | night_audit.php | Night Auditor, Manager     | Simulation run, folio balance, discrepancy report |
| Reports         | reports.php     | Manager, Accountant        | Revenue, HK performance, guest LTV charts         |
| Audit Trail     | audit_trail.php | Admin, Auditor             | Immutable log viewer, filter by date/action/user  |

## Appendix B — Requirements Traceability Matrix

| UC-ID | Activity Diagram | Sequence Diagram | Collaboration | Class(es)             | DB Table(s)             | PHP File         | Test Cases   |
|-------|------------------|------------------|---------------|-----------------------|-------------------------|------------------|--------------|
| UC-01 | AD-01            | SD-01            | CD-01         | Room, AllocEngine     | rooms, reservations     | reservations.php | WB-03, BB-02 |
| UC-02 | AD-02            | SD-07            | CD-07         | Reservation, AuditLog | reservations, state_log | reservations.php | WB-02, BB-07 |
| UC-05 | AD-06            | SD-05            | CD-05         | PricingEngine         | reservations            | checkin.php      | WB-05        |
| UC-14 | —                | SD-14            | CD-14         | Guest, LoyaltyAcct    | guests                  | dashboard.php    | BB-VIP       |
| UC-19 | —                | SD-19            | CD-19         | Guest, AuditLog       | guests, audit_logs      | guests.php       | BB-GDPR      |
| UC-23 | AD-05            | SD-23            | CD-23         | Room (Observer)       | rooms, hk_tasks         | housekeeping.php | BB-HK        |
| UC-32 | AD-02            | SD-32            | CD-32         | Room, AlertService    | hk_tasks                | housekeeping.php | WB-01        |
| UC-33 | —                | SD-33            | CD-33         | Folio, Payment, Tax   | folios, payments        | folios.php       | WB-04, BB-03 |
| UC-36 | AD-04            | SD-36            | CD-36         | Payment, Folio        | payments, folios        | folios.php       | BB-03        |

| UC-ID | Activity Diagram | Sequence Diagram | Collaboration | Class(es)          | DB Table(s)          | PHP File        | Test Cases   |
|-------|------------------|------------------|---------------|--------------------|----------------------|-----------------|--------------|
| UC-40 | —                | SD-40            | CD-40         | Staff, Role, Auth  | staff, audit_logs    | staff.php       | WB-06        |
| UC-41 | —                | —                | —             | AuditLog           | audit_logs           | audit_trail.php | NFR-08       |
| UC-42 | AD-03            | SD-05            | CD-05         | NightAudit, Folio  | night_audits         | night_audit.php | WB-05, BB-06 |
| UC-25 | —                | SD-25            | CD-25         | MaintRequest, Room | maintenance_requests | maintenance.php | BB-MNT       |